

Netflix Recommendation System

using Collaborative Filtering and Matrix Factorization (SVD)

Personalized Movie Recommendation using Collaborative Filtering and Matrix Factorization



Machine Learning

Collaborative Filtering & SVD



Recommendation System

Personalized Movie Suggestions

Project Team

Snehitha Jarpala (23323016)

Pranathi Kokkiligadda (23323019)



Problem Overview



The Problem

- Thousands of movies make discovery difficult
- Users have diverse viewing preferences
- Manual browsing reduces engagement



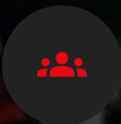
The Objective

- Build a personalized recommendation engine
- Learn preferences from rating history
- Recommend Top-N relevant movies



Why It Matters

- Improves user experience
- Increases content consumption
- Enhances user retention
- Enables personalized entertainment



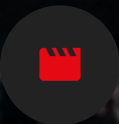
Users



Movie Ratings



Recommendation Engine



Personalized
Suggestions

- 100K+ Ratings
- 9K+ Movies
- 671 Users



Key Insight: SVD learns hidden user preferences from historical ratings and generates personalized recommendations with higher accuracy.

Dataset and Methodology

Dataset Statistics

225

Movies

283,705

Users

1,000,000

Ratings

Methodology Pipeline



Algorithms Used: Item-Based Collaborative Filtering | Singular Value Decomposition (SVD)

Key Insights from Data Analysis

Key Findings

- Most ratings are between 3 and 5 stars.
- Highly active users contribute most ratings.
- Popular movies receive significantly more interactions.
- User-Movie interaction matrix is sparse.

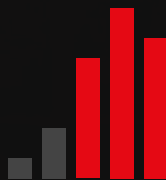
Sparsity Metric

98.43% Sparse Matrix

Most user-movie interactions are missing, making recommendation challenging.

"User preferences are highly personalized, making recommendation systems essential."

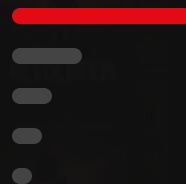
Rating Distribution



Movie Popularity



User Activity



High sparsity and personalized viewing behavior justify the use of Collaborative Filtering and SVD-based recommendation models.

Model Comparison & Selection

✓ Final Selected Model: SVD

Item-Based Collaborative Filtering

- Uses movie-to-movie similarity
- Recommends similar movies
- Simple and interpretable

Singular Value Decomposition (SVD)

- Matrix factorization technique
- Learns hidden user preferences
- Handles sparse data effectively
- Produces better recommendations

Input Matrix
User-Movie Ratings


Factorization

Latent Factors
Hidden Features

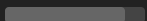
Experimental Results

Comparison of Item-Based Collaborative Filtering and SVD

RMSE ↓

Item-CF

1.1421



SVD

0.9924

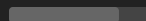


WINNER: SVD

MAE ↓

Item-CF

0.8870



SVD

0.7775

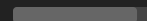


WINNER: SVD

MAP@10 ↑

Item-CF

0.9372



SVD

0.9512



WINNER: SVD

Key Finding: SVD consistently outperformed Item-Based Collaborative Filtering across all evaluation metrics.



Selected Model: SVD

- **Lowest RMSE (0.9924)**
- **Lowest MAE (0.7775)**
- **Highest MAP@10 (0.9512)**

Recommendation Examples

Sample Recommendations Generated by SVD Model

Top-10 Personalized Recommendations



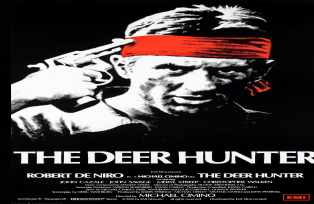
The Game



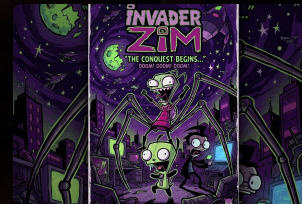
Lilo and Stitch



Reservoir Dogs



The Deer Hunter



Invader Zim



X2: X-Men United



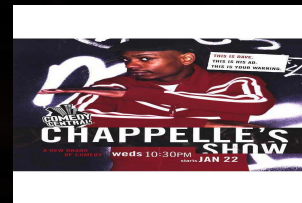
ST: Deep Space Nine



Mostly Martha



The Chorus



Chappelle's Show

These recommendations were generated using Singular Value Decomposition (SVD) based on user-rating patterns and latent preference learning.

Conclusion & Future Work

Key Conclusions

- Developed a personalized movie recommendation system using collaborative filtering techniques.
- Implemented and evaluated Item-Based Collaborative Filtering and Matrix Factorization (SVD).
- SVD achieved the best performance across RMSE, MAE, and MAP@10 metrics.
- Built an interactive Streamlit dashboard for recommendation visualization.

Project Outcome

SVD was selected as the final recommendation model due to superior accuracy, ranking performance, and ability to capture hidden user preferences.

Future Work

- Hybrid recommendation systems
- Genre-aware recommendations
- Cold-start problem handling
- Deep learning recommendation models
- Cloud deployment and scalability

"SVD effectively captures hidden user preferences, delivering more accurate and personalized movie recommendations."

THANK YOU!

Questions & Discussion

